Defense vs Tech Summary Analysis Report

Dataset and source

This project uses daily adjusted closing prices for Lockheed Martin (LMT), Northrop Grumman (NOC), General Dynamics (GD), and the Technology Select Sector SPDR Fund (XLK). The data was downloaded from Yahoo Finance using the yfinance package. The sample runs from January 2020 through December 2025. Adjusted close prices were used so return comparisons reflect investor-relevant price adjustments. Missing values were checked first, then forward-filled only to keep the time series aligned for rolling calculations.

Analysis methods

The notebook used normalized price performance, daily return distributions, 30-day rolling volatility, 30-day rolling correlation, a GARCH conditional volatility model, pair regressions against XLK, and a small regularization comparison using OLS, Ridge, and Lasso. It also tested a regime strategy twice: once in a biased same-day form and once in a corrected lagged form. This was useful because it showed how an apparently strong strategy can weaken once the timing is made realistic.

Key findings

• XLK had the strongest long-run performance in the sample, while GD had the lowest annualized volatility.

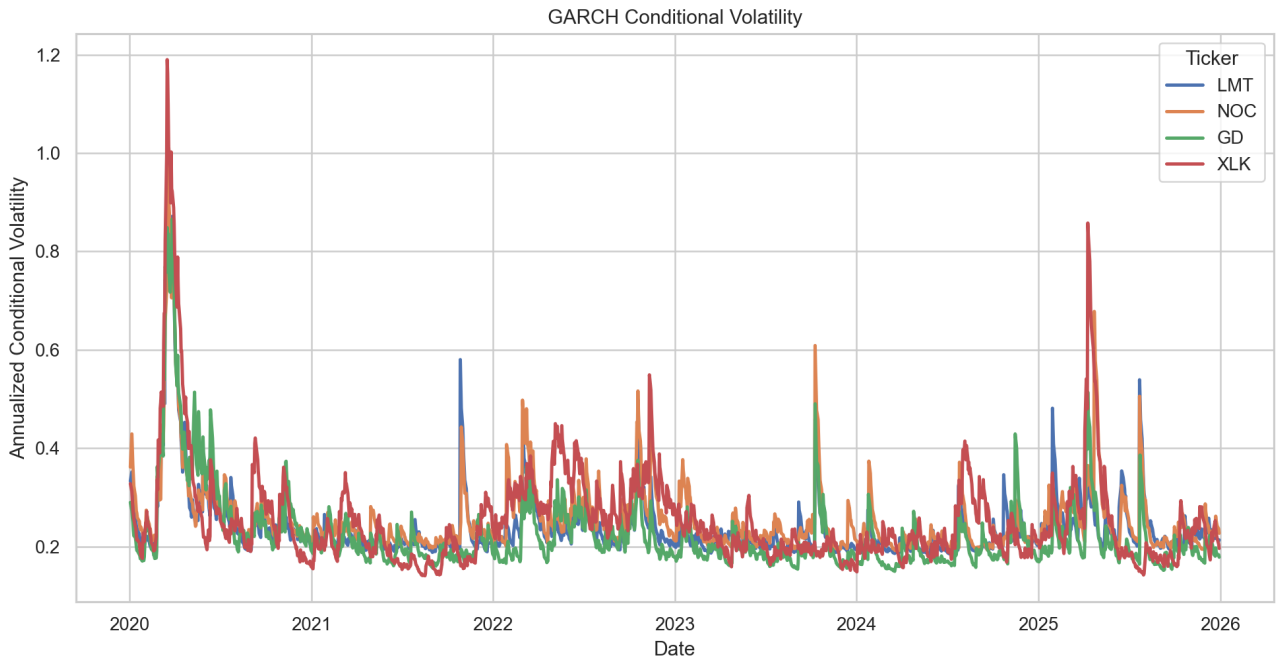
• NOC showed the weakest full-period correlation to XLK, so it was the best candidate for conditional diversification.

• GD had the strongest regression fit to XLK, which means it behaved most like a tech-sensitive equity among the defense names.

• LMT had the highest kurtosis, meaning its return distribution had the fattest tails and more extreme moves.

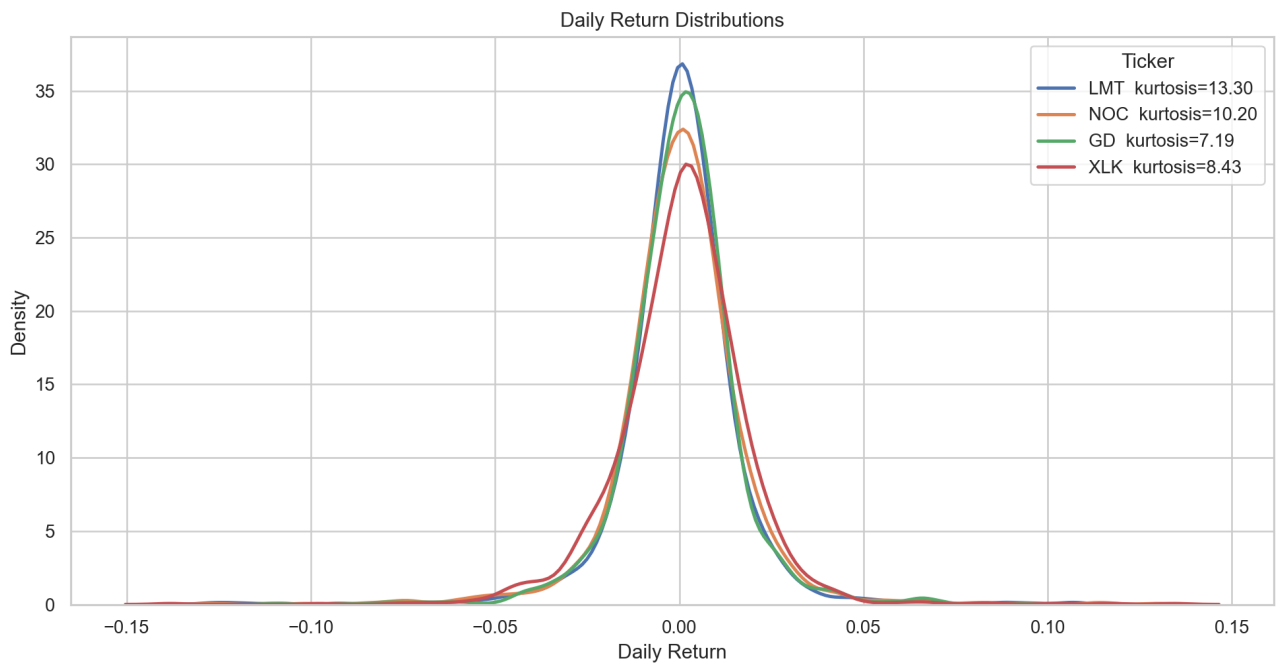
• The strategy tests mattered less as trade ideas and more as diagnostics. The same-day version looked strong, but the lagged version lost much of that edge once lookahead bias was removed.

Visual 1: garch conditional volatility



This chart shows model-based conditional volatility rather than a simple rolling standard deviation. In plain terms, it is the model's running estimate of how unstable each asset is after taking recent shocks into account. The main takeaway is that volatility came in clusters rather than staying constant. XLK generally had the most persistent high-volatility bursts, while the defense names usually looked calmer but still spiked during stress periods.

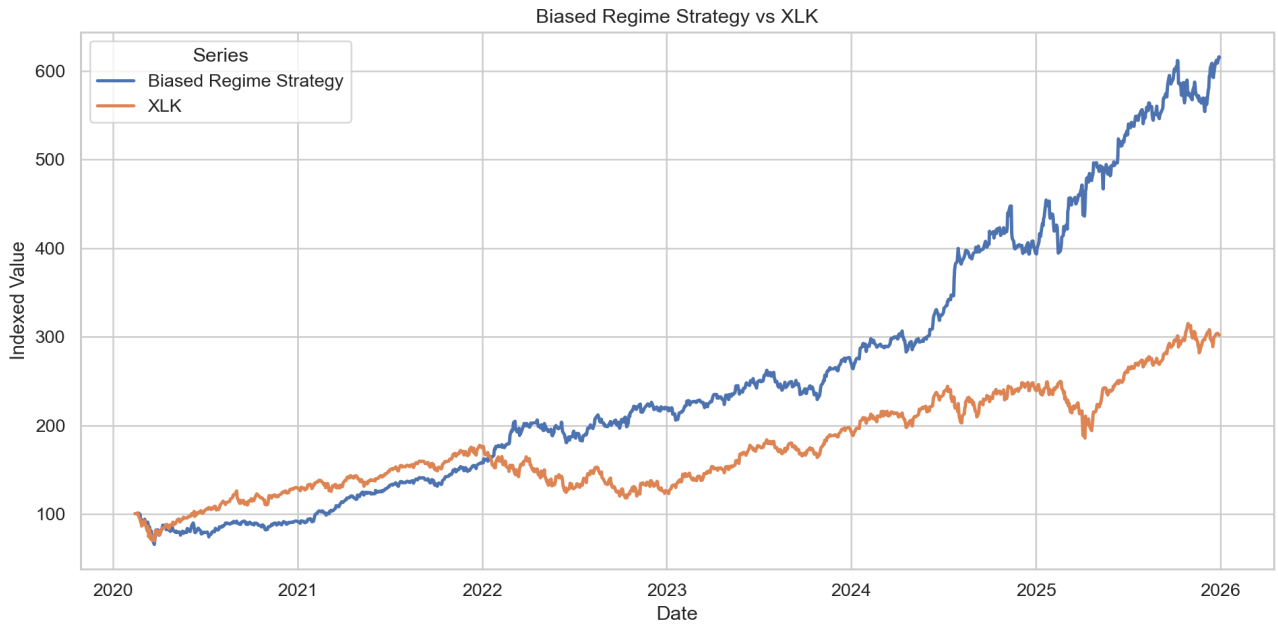
Visual 2: daily return distributions



This chart compares the shape of daily returns for each asset. Kurtosis helps explain tail risk: higher kurtosis means more weight in the tails and therefore a greater chance of unusually

large daily moves. In this sample, the return curves showed that the assets were not just different in average return; they were also different in how often they produced outsized moves. That matters for risk even when average performance looks acceptable.

Visual 3: biased regime strategy



The biased regime strategy ended near 614.8 versus XLK near 301.6. At first glance, that looks like strong evidence that switching between XLK and defense stocks added value. The problem is that this version used same-day information to make same-day decisions. That means the backtest was unintentionally allowed to react to information it would not have known in real time.

Visual 4: lagged regime strategy

Once the signal was lagged by one trading day, the regime strategy ended near 224.7 versus XLK near 300.3. This corrected version is the more credible result. Its weaker performance

implies that the earlier outperformance was mostly a false positive caused by lookahead bias. This is an important practical lesson: intuitive timing rules can look powerful until the implementation is made realistic.

Brief reflection

The clearest conclusion is not that a defense rotation strategy beat tech. Instead, the evidence suggests that defense stocks were only partial and unstable diversifiers versus XLK. Some relationships weakened in certain windows, but they were not stable enough to treat defense as a constant hedge. The project was still useful because it showed how to clean financial time series, compare risk patterns, and test whether a promising chart pattern survives a more realistic backtest.